Alexandros Mylonakis

Ioannina, Greece

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Professional Summary

AI Engineer at NIKI Digital Engineering, onboarding onto engineering simulation and web-application projects. Integrated Master's graduate in Computer Science & Engineering (July 2026), with thesis research on gender bias in LLM embeddings and hands-on Python and NLP experience.

Education

University of Ioannina **July 2026**

Diploma in Computer Science & Engineering (Integrated Master) *Ioannina, Greece*

Experience

NIKI Digital Engineering **Sep. 2026 – Present**

AI Engineer *Ioannina, Greece (On-Site)*

- Onboarding onto client projects in engineering simulation and web applications, through code reading, shadowing and hands-on exploration of the project environments.
- Building working knowledge of MATLAB, TypeScript, React and JSON while collaborating with engineers across Niki and partner companies.
- Assigned to an upcoming project developing an AI-agent-based solution.

Ontovisual **May 2026 – June 2026**

Software Engineering Intern *On-Site / Ioannina, Greece*

- Built an automated typesetting engine using Shape Grammars and Guillotine Partitioning to map generative AI output onto multi-page, magazine-style layouts.
- Constrained LLM output to strictly formatted JSON through structured prompts and a client-side validation schema, feeding a deserialization pipeline with overflow and look-ahead buffering.
- Added typography guardrails that resolve Greek text-wrapping collisions by truncating text in edit mode.

Technical Projects

Master's Thesis: Measuring Bias in LLM2Vec Representations **Feb. 2025 – Feb. 2026**

Python, PyTorch, LLMs, NLP LLM2Vec_Thesis

- Developed an automated diagnostic framework to benchmark 6 public LLM2Vec-adapted bidirectional encoders against traditional Word2Vec and BERT baselines.
- Built gender-bias axes using L2-normalization and PCA; explained-variance analysis and comparison with single-pair (he–she) axes indicated that distributional methods are more stable for measuring bias in hidden states.
- Compared model sizes and architectures, observing that the structurally pruned Sheared-LLaMA-1.3B showed lower neutral-occupation bias than larger models, suggesting pruning may act as a denoising step.

Additional Experience

PanCar Rentals **2015 – 2022**

Fleet Operations & Client Service *Chania, Crete*

- Utilized strategic communication to de-escalate and resolve last-minute scheduling conflicts under strict time constraints, maintaining high customer satisfaction across 8 consecutive seasons.
- Demonstrated rapid adaptability in a fast-paced operational hub by onboarding new seasonal staff and adjusting fleet logistics to meet shifting daily demands.

University of Ioannina Radio Club **2017 – 2021**

IT Support & Web Administrator *Ioannina, Greece*

- Fostered effective teamwork by coordinating technical setups and logistics alongside cross-functional teams to deliver successful live music festivals and university seminars.
- Maintained and updated the station's digital infrastructure and website content, ensuring high uptime and accessibility for listeners.

Technical Skills

Languages: Greek (Native), English (C2 Certified - Michigan ECPE)

Programming Languages: Python, Java, C, JavaScript, TypeScript, MATLAB, HTML/CSS, SQL
ML & NLP: PyTorch, Hugging Face, Scikit-learn, NumPy, Pandas, Prompt Engineering
Frameworks & Web: React, Redux, craft.js
Data Tools: Databricks and Power BI (introductory)
Developer Tools: Git, VS Code, Linux, Jupyter Notebooks, Claude Code, Antigravity